

Project Name: SplitsFromVideo
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1.1 System Architecture Diagram

User Types	User-Facing Machines	Non-User-Facing Machines
Speedrunners	User Desktop	yt-dlp
Speedrun data analyst		cap-from-youtube
		PaddleOCR
		PARTS

1.2 Machine and Software Component Mapping

User-Facing Machines:

- User Desktop:
Users will have a gui application with the ability to input a YouTube link to a video that contains a speedrun with a livesplit window within it. The video will appear to the user, where the user will define the area of the split timer, and an option to choose which format they would like their output data to be in and a textbox for what they would like their data to be called. The option to download that data will appear and it will begin the OCR process of extracting the split data.

Non-User-Facing Machines:

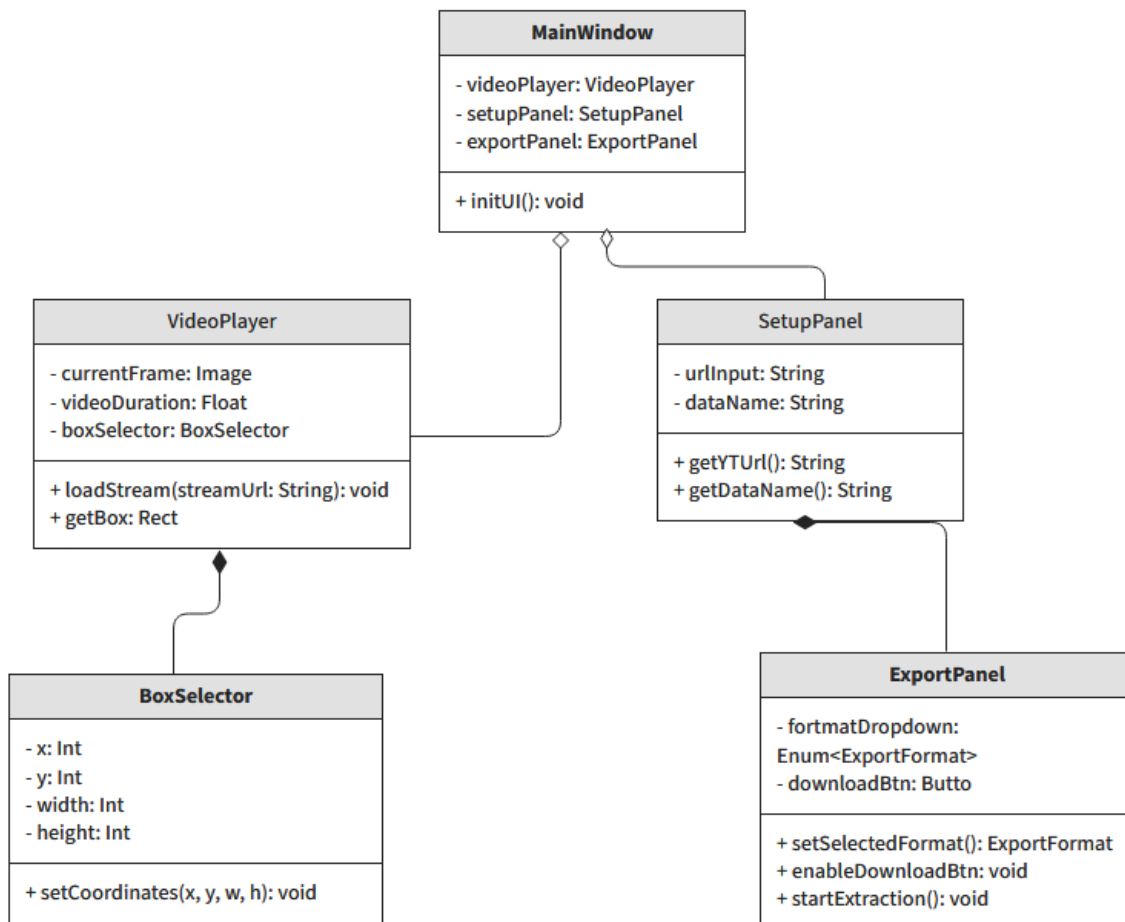
- yt-dlp:
This process will connect to youtube and use the link given by the user to fetch a video stream and download it to the user's storage. This is used to have a stream for the user to scroll through the video and find a good frame to draw the timer boundary for PaddleOCR to use.
- cap-from-youtube:
This process allows the program to only download needed frames from the given YouTube video, which will be used by PaddleOCR combined with the restricted border for the timer to only read the necessary parts of the video.

- **PaddleOCR:**
This process will examine the selected part of the stream given from cap-from-youtube, defined by the user within the app's gui, and will read the timer's time. It will then record the time in the format requested by the user.
- **PARTS:**
This process will search the video backwards, finding the first instance of the timer being stopped, then going back by the given time on the timer, going back a single frame to the next time it would have a time, and repeating the process until the beginning of the speedrun. This method should save the user from needing more resources than necessary, while speeding up the search process.

2.1 Class / Module Breakdown

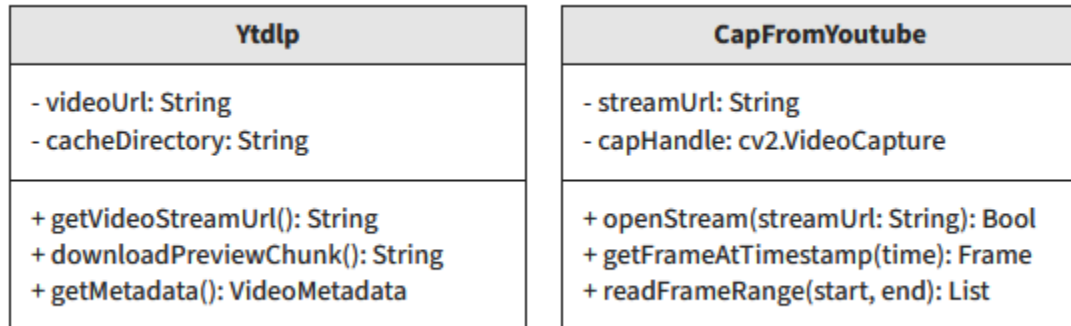
User-Facing Machines:

- User Desktop:

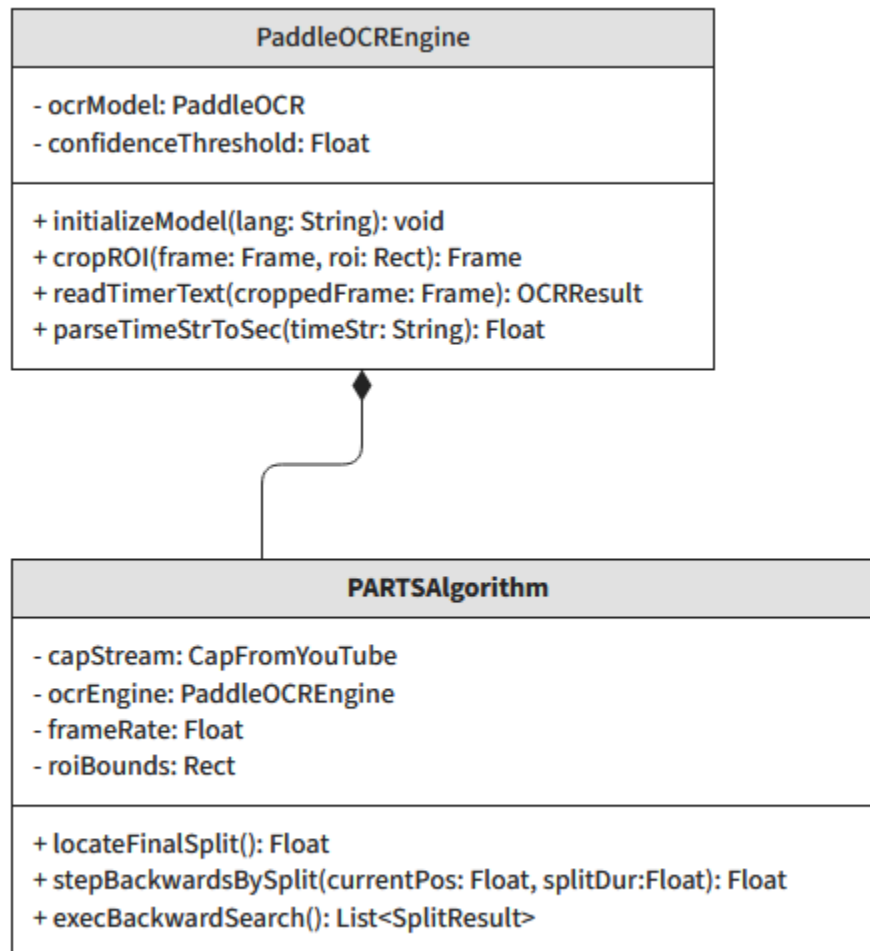


Non-User-Facing Machines:

- ytdlp + capfromyoutube



- PaddleOCR + PARTS



3.1 Graphical User Interface Design

About

SplitsFromVideo

How to use

Settings

Input YouTube URL:



Format Select ▾

File name input

Download

Splits:

4:30.94
4:18.51
3:08.33
2:28.03
4:15.70
5:35.46
3:01.97
2:36.64
3:08.14
3:34.57
2:55.49
5:35.87
2:50.45
2:56.54
1:39.33
2:38.28
5:37.55
2:03.08